



Birla Vishvakarma Mahavidyalaya Engineering College
(An Autonomous Institution)
Managed by Charutar Vidya Mandal
IT Department



Project Definition Approval Presentation
Project (4IT31)
Academic Year 2026-27

Group 09

Preference 1: Chess Playing Robotic Arm

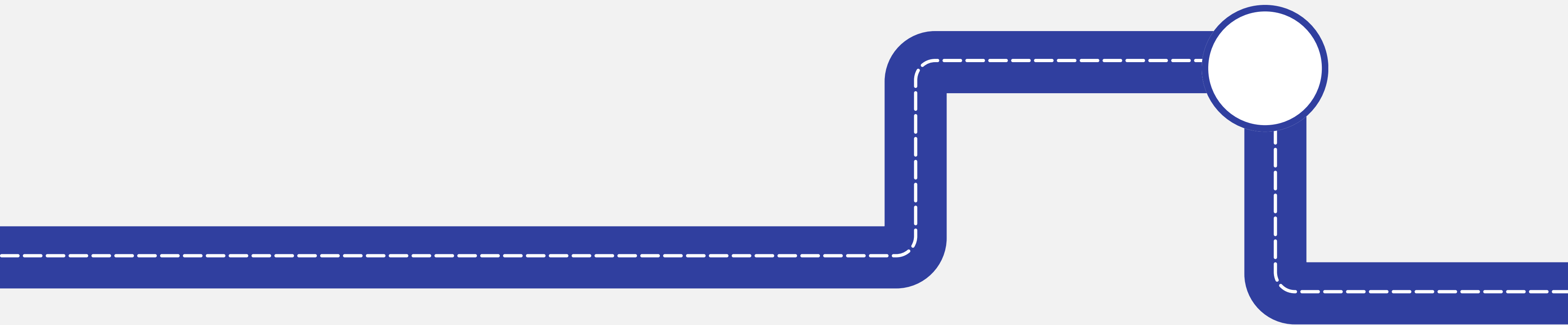
Preference 2: Smart Finance Planner

Team:

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Chess Playing Robotic Arm





Project Definition and Objectives

Definition

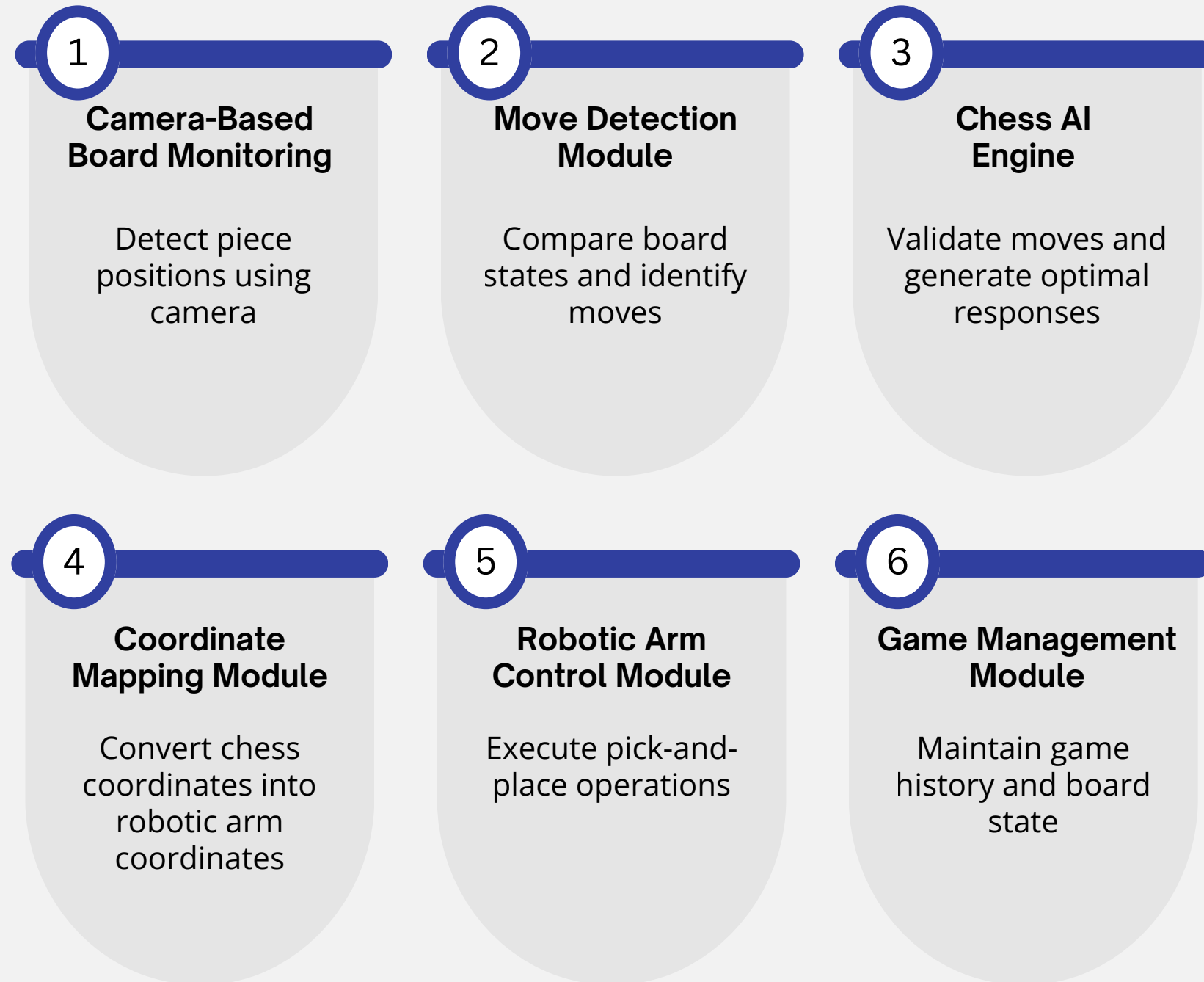
A smart chess-playing robotic system that uses computer vision to detect piece movements. The system compares previous and current board states, validates moves using a chess engine, and controls a robotic arm to perform the AI-generated move on the physical board.

Objectives

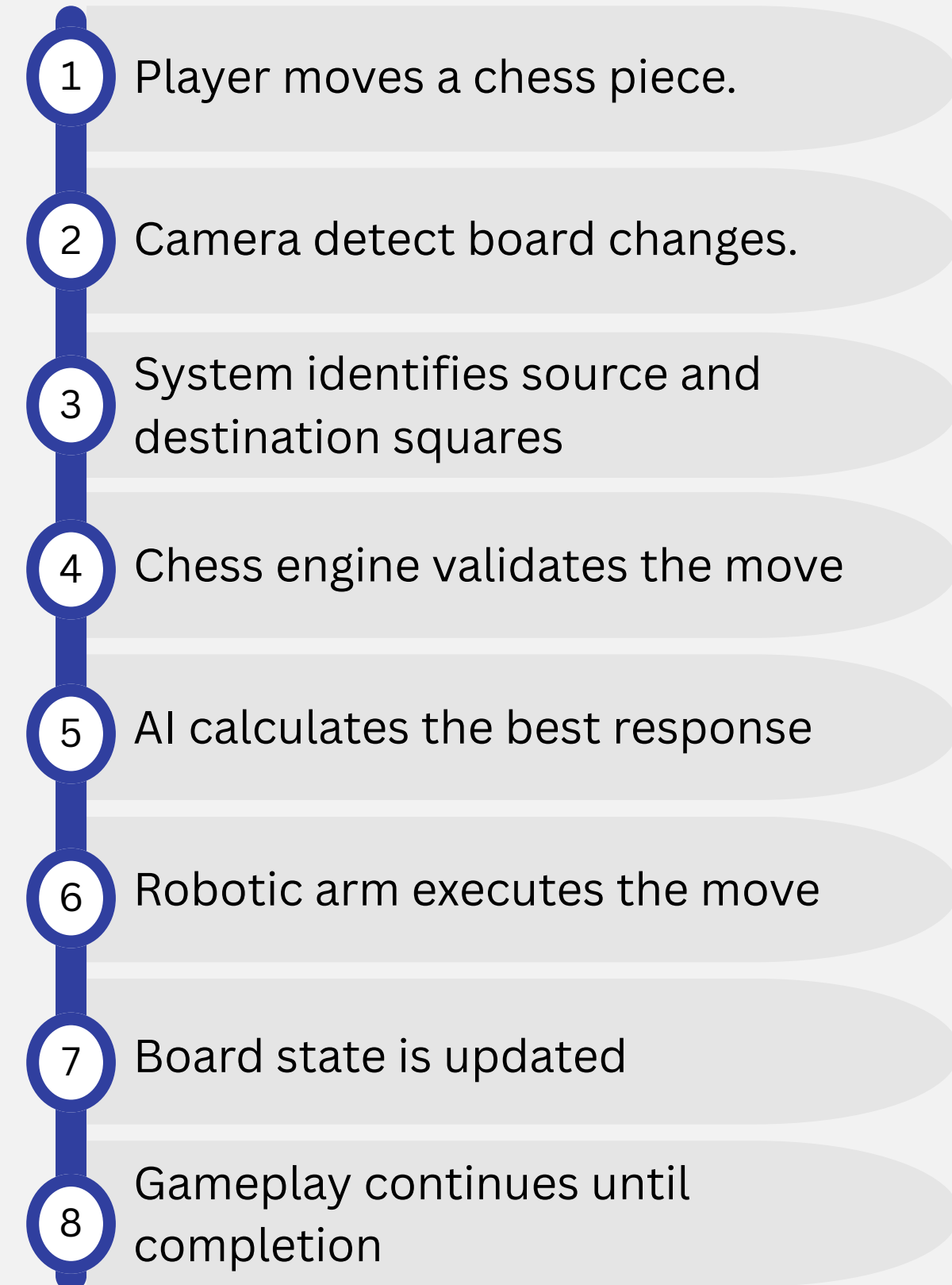
- Automate physical chess gameplay and help remote chess players.
- Detect chess moves using camera.
- Integrate AI for intelligent move generation.
- Control a robotic arm for piece movement.
- Demonstrate the application of AI and robotics in board games.



Modules



System Functionality





Feasibility

1

Technical Feasibility

Uses Raspberry Pi, magnetic sensors, robotic arm, Python, and Stockfish AI with readily available components

2

Economic Feasibility

Moderate cost; hardware components are affordable and easily available online

3

Operational Feasibility

Fully automated system with reliable sensor-based move detection and robotic piece movement

4

Future Feasibility

Can be extended with online multiplayer, voice control, automatic board setup,

Research Findings

1

Automated Chess Gameplay With Computer Vision and A Robotic Arm

2

Development of an autonomous chess robot system using computer vision and deep learning

3

Chess Playing Robotic Arm

4

Robotic arm playing chess

Tech Stack

Front End	  
Back End	 
AI Engine	
Database	
Hardware	 
Dataset	<u>Kaggle Chess Dataset</u>
Tools	 

Expected Outcomes

- Accurate move detection
- Autonomous chess gameplay
- Intelligent AI responses.
- Reliable robotic arm movement

Scope

- Human vs AI gameplay
- Educational robotics platform
- Smart board game automation
- Future support for online and remote chess

Smart Finance Planner

The background features several decorative blue elements. In the top-left corner, there is a solid blue shape with a wavy, organic edge. A solid blue horizontal line is positioned below the title. A thick blue line with rounded corners runs horizontally across the lower half of the image, featuring two white circular nodes. A dashed white line follows the path of this thick blue line. On the right side, a vertical blue line with rounded corners runs from the top to the middle of the frame, also featuring a dashed white line.



Project Definition and Objectives

Definition

An AI-assisted financial planning system designed to help users manage daily expenses, utility bills, subscriptions, vehicle expenses, savings, and budgets from a single platform. The system provides spending analysis, reminders, and personalized financial insights to improve financial management

Objectives

- Maintain a centralized record of all expenses.
- Track recurring monthly bills.
- Categorize and analyze spending patterns.
- Provide budget recommendations.
- Generate financial reports and insights.
- Improve financial awareness and planning



Modules

1

Expense Tracking Module

Record and categorize expenses.
Manage import records

2

Bill Management Module

Electricity, gas bills, vehicle expenses, rent and subscriptions due date tracking.

3

Budget Planning Module

Monthly budget creation and budget monitoring

4

Analytics and Reporting Module

Expense summaries, Spending trends. Monthly and yearly reports, Graphical dashboards.

5

AI Recommendation Module

Budget suggestions, Expense forecasting, Unusual spending detection.

6

Notification Module

Reminders and budget alerts

System Functionality

1

User logs income and expenses

2

Expenses are categorized automatically or manually

3

Bills and recurring expenses are recorded

4

System monitors spending against budgets.

5

Reports and analytics are generated

6

AI predicts future expenses

7

Users receive reminders and recommendations

8

Financial overview on dashboard



Feasibility

1

Technical Feasibility

Developed using Python, FastAPI, MySQL, and AI libraries with standard development tools

2

Economic Feasibility

Low-cost implementation using open-source software; no specialized hardware required

3

Operational Feasibility

User-friendly platform for managing expenses, bills, budgets, and financial insights

4

Future Feasibility

Can be expanded with banking integration, OCR-based bill scanning, mobile application, and AI financial assistant

Research Findings

1

Financial Planning Tools

2

Artificial Intelligence and Consumer Financial Behavior: A Systematic Literature Review and Agenda for Future Research

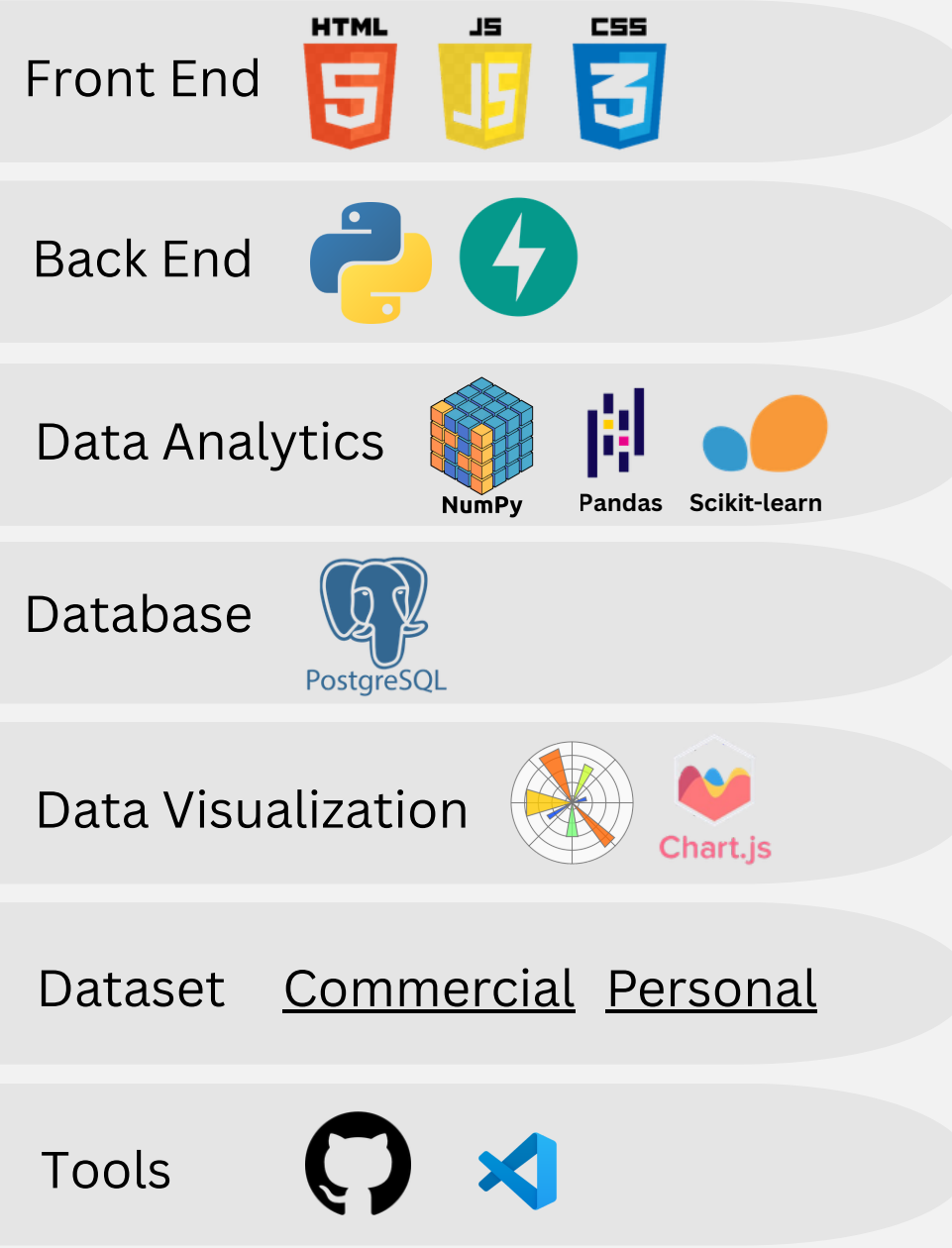
3

AI-Powered Personal Finance Tracker

4

AI-Driven Personal Finance Management System

Tech Stack



Expected Outcomes

- Centralized expense management.
- Better financial planning.
- Reduced missed bill payments.
- Personalized budgeting recommendations.

Scope

- Personal and family finance tracking.
- Small business expense monitoring.
- Future integration with banking systems.
- Investment and tax planning support.

THANK YOU
